HOLY CROSS COLLEGE

Life to the Full

12 PHYSICS

SPECIAL RELATIVITY

RELATIVITY

Moving Through Spacetime

People are very used to moving through time and space. They are moving through time at the rate of 24 hours every day, but are not moving through space (unless they are in a train).

If a person takes a fast trip to the other side of the world, they have moved through space as well as time.

Einstein's radical idea was that ***travel through space and travel through time are interrelated.*** In a sense, ***the faster a person travels through space, the less they travel through time***. It is as though the person moves through what Einstein called ***spacetime*** at a constant rate, but the amount of space and the amount of time depend on who measures them.

If someone on Earth observes a space traveller, it turns out that the measurements of her time elapsed and distance travelled do not agree with her measurements. Her time is going ***more slowly*** than ours. However, the traveller sees that our clocks are going slow. Who is right?

The answer is that ***both*** are. Time and space are ***relative***. But that is what Einstein's theory of relativity is all about.

Frames of reference

Velocity is always measured relative to some particular co-ordinate system or frame of reference.

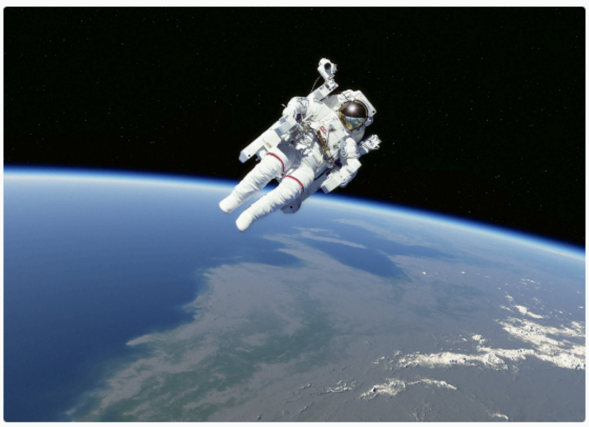
If you make measurements from a frame of reference that is moving at a ***constant velocity***, then any measurements of change in velocity will agree with those made by observers in any other constant-velocity frame of reference.

Frames of reference that move with a constant velocity are called ***inertial frames of reference***.

Example

> A car is moving at 60 kmh^{-1} relative to the ground. The assumption is that the car's velocity is measured by a radar gun that is at rest relative to the road.
>
> If the reading is made by a radar gun mounted on a police car following behind at 50 kmh^{-1} relative to the road, the same car is only travelling at 10 kmh^{-1}.

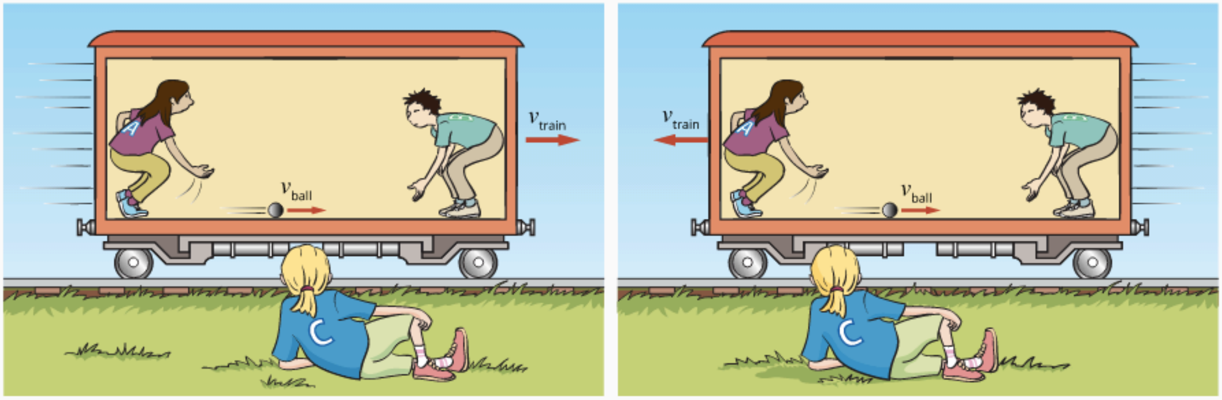
According to Galileo's equations and Newton's laws, ***the velocity of objects can be calculated relative to any frame of reference as long as the velocities of the frames of reference are known***.



8.1.4 • This astronaut moves slowly relative to the space shuttle (from where this photograph was taken), at a speed of nearly 8000 m s^{-1} relative to observers on Earth, and at a speed of about 30 000 m s^{-1} in the frame of reference of the Sun.

Example

For example, if Amaya and Binh are on a train, and they roll a ball towards the front of the train at $2\ \mathrm{m\ s^{-1}}$, then relative to the train the ball is moving at $2\ \mathrm{m\ s^{-1}}$ forwards. If Clare is standing still on a platform and sees Binh rolling the ball, then the velocity of the ball relative to Clare depends on the total of the velocity of the train plus the velocity of the ball. If the train is moving at $10\ \mathrm{m\ s^{-1}}$ forwards, then Clare will see the ball moving at $12\ \mathrm{m\ s^{-1}}$ forwards $(10+2=12)$. If the train is moving backwards at $10\ \mathrm{m\ s^{-1}}$, then Clare will see the ball move at $8\ \mathrm{m\ s^{-1}}$ backwards $(-10+2=-8)$, and if the train is stationary, then Clare will see the ball moving at $2\ \mathrm{m\ s^{-1}}$ forwards, just as Amaya and Binh do $(0+2=2)$.



8.1.5 • Relative to a stationary observer in a frame of reference, the overall velocity of an object moving in a moving frame of reference will depend on the velocity of the object within the moving frame and the velocity of the moving frame with respect to the observer's frame.

SPEED OF LIGHT

The speed of light is one of the most accurately measured constants in physics and its value has been determined very precisely by many different methods.

The accepted value of the speed of light in a vacuum is now:

$$\boxed{c = 2.99792458\ \mathrm{x}10^{8}\ \mathrm{ms^{-1}}}$$

Physics file: The speed of light

The value of $c = 2.997\ 924\ 58 \times 10^{8}\ \mathrm{m\ s^{-1}}$ for the speed of light is the accepted value, and is now used to define other standards. The speed of light is now the **primary standard** with the unit of length, the metre, being defined as the distance light travels in a time equal to $\frac{1}{2.997\ 924\ 58 \times 10^{8}}$ seconds.

This value is consistent with the old standard –the length between two marks on a special bar kept at Sèvres, near Paris. In fact, the distance between the marks cannot be measured to this degree of accuracy.

James Maxwell, working in 1864, derived the speed of light from a mathematical analysis of the propagation of electromagnetic waves.

He took the concept of electric and magnetic fields and worked with them to produce a mathematical description that encompassed all known electromagnetic phenomena. This description can be summed up in four famous equations, referred to as Maxwell's equations.

$$\oint \vec{E} \cdot d\vec{A} = q/\epsilon_0$$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i$$

8.1.9 • Maxwell's equations: The first describes the electric field around a charge in a vacuum. The second describes the fact that magnetic fields are continuous. The third is Faraday's law of electromagnetic induction, which tells you that a magnetic field is generated by a changing magnetic flux. The fourth tells you that a magnetic field is generated by electric currents and/or changing electric flux.

He found that his equations could be used to predict that changing perpendicular electric and magnetic fields could self-propagate through space at a speed given by the fundamental electric and magnetic force constants.

$$\boxed{c = \frac{1}{\sqrt{\varepsilon_o \mu_o}}}$$

where ε_o = the electric permittivity of a vacuum.
μ_o = the magnetic permeability of a vacuum.

This speed turned out to be 3.00×10^8 ms^{-1}

Maxwell faced a conundrum, however. His derivation of this speed showed that its value depended only on the two constants for the strength of the electric and magnetic force fields near charges or currents. There was no indication that it should be any different if, for example, the source of the light was in motion.

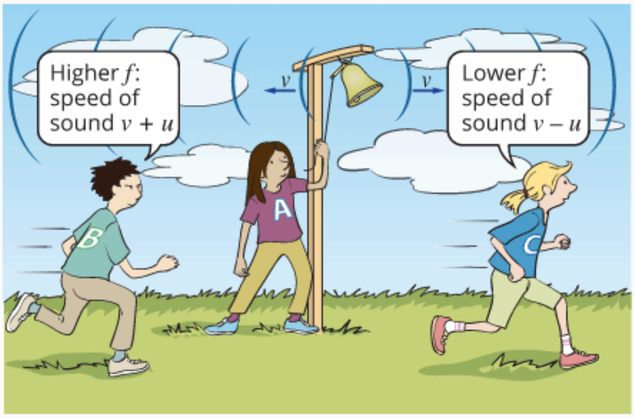
i.e. ***Speed of light is constant in all reference frames***.

There was no indication that it should be any different if the source of the light was in motion.

What was surprising, though, was that there appeared to be no allowance for the speed of the 'receiver' of the light.

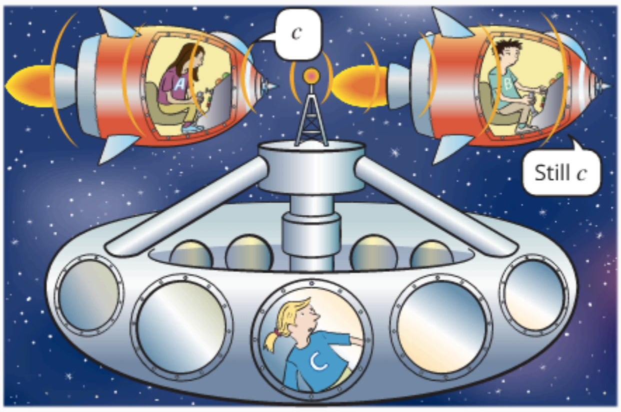
e.g. If you run towards a ball thrown at you, the speed of the ball, relative to you, is greater than if you stand still.

Similarly, if a person is moving towards a source of sound, the speed of the sound (the product of the measured frequency and wavelength) will be the sum of the speed of sound in the medium and the speed of the person.



8.1.10 • As Binh runs towards Amaya, who is ringing the bell, the speed of sound will seem higher than normal. As Clare runs away from Amaya, the speed of sound will seem lower. In both cases, the speed is equal to $f\lambda$, where the wavelength remains the same. However, in the first scenario the frequency will seem higher, and in the second it will seem lower.

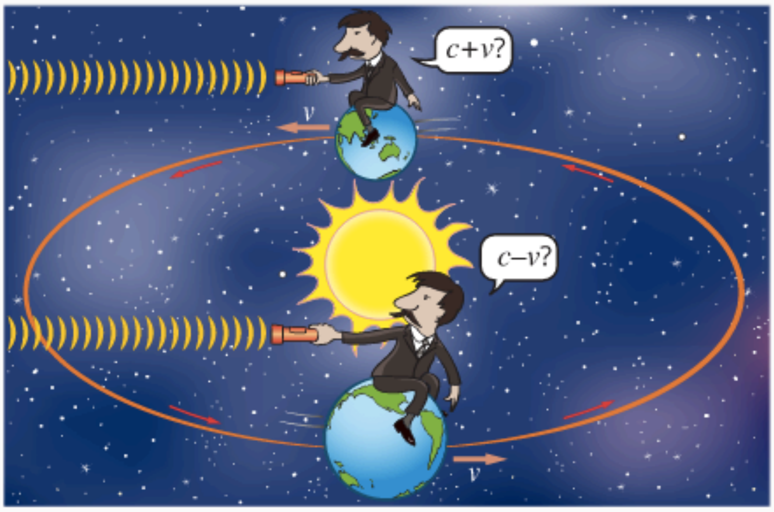
Maxwell's equations refused to make any allowance for the motion of the observer of light waves. Whatever the speed of the observer, the measured speed of light is 3.00×10^8 ms^{-1} - something quite against all the known laws of physics.



8.1.11 • As Amaya speeds towards Clare's space station and its beacon light, she measures the apparent speed of light to be c. Maxwell's equations state that Binh, as he moves away from the space station, will also find that the speed of light is exactly c.

The Michelson-Morley Experiment

In the 1880's, Albert Michelson and Edward Morley set up a device known as an interferometer, which was able to measure the very small differences in the time taken for light to travel in two mutually perpendicular directions. They were able to rotate the whole apparatus and hoped to detect the small difference that should result from the fact that one of the directions was to be the same as that in which the Earth was travelling and the other at right angles. However, ***they found no difference***.



8.1.12 • The basic principle of the Michelson–Morley experiment. If the aether is fixed relative to the Sun, and the light is travelling (at c relative to the aether) in the same direction as the Earth, the apparent speed should be less than c. Six months later, light travelling in the same direction should appear faster than c.

Six months later, when the Earth would have to be travelling in the opposite direction relative to the light coming from the Sun, there was still no difference in the measured speeds.

EINSTEIN

Einstein was the archetypal theoretician - the only significant experiments he ever did were ***thought experiments***, or 'Gedanken' as they are called in his first language (German). Many of his Gedanken experiments involved thinking of situations that involved ***two frames of reference moving with a steady relative velocity***. Newton had referred to these as ***inertial frames of reference***, as the law of inertia applied within them, but ***not within frames that were accelerating***.

He stated that all inertial frames of reference must be equally valid, and that the laws of physics must apply equally in any frame of reference that is moving at a constant velocity. A consequence of this is that there is no physics experiment you can do, completely within a particular frame of reference, to tell that you are moving.

e.g. As you speed along in a Gedanken train with the blinds down, you cannot measure your speed, at least not without peeking out the window, or connecting something to the wheels, which are in contact with the outside frame of reference.

You can tell if you are accelerating easily enough: just hang a pendulum from the ceiling. However, the pendulum will hang straight down if your velocity is constant, whether you are travelling steadily at 100 kmh^{-1} or are stopped in the station.

THE SPECIAL THEORY OF RELATIVITY

Paradox - how could two observers travelling at different speeds see the same light beam travelling at the same speed?

Einstein put forward two simple postulates.

- The principle of relativity states that no law of physics can identify a state of absolute rest. (In other words, no single inertial frame of reference is better than any other.)
- The speed of light will always be the same no matter what the motion of the light source or observer.

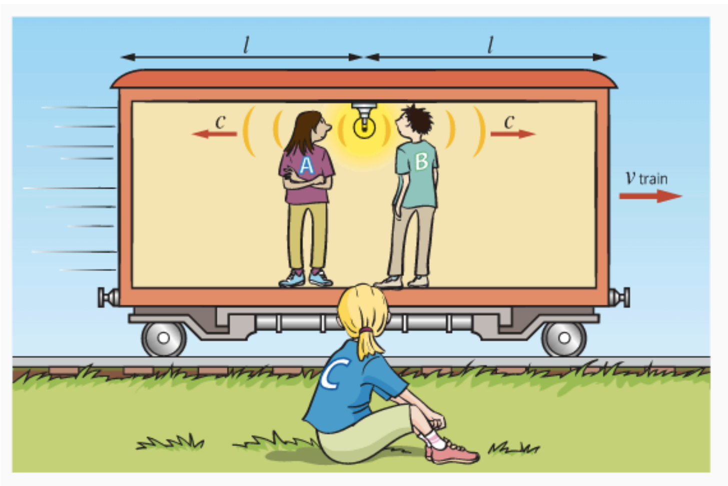
Newton's laws are based on the idea that space and time are how they seem to a person on Earth - constant, uniform and straight. Space is like a big set *xyz* axes that always remain mutually perpendicular, and in which distances can be calculated exactly according to Pythagoras's rule. In this space, ***time flows on at a constant rate that is the same everywhere***.

Einstein realised that the assumptions Newton made may not in fact be valid, at least not on scales involving huge distances and speeds approaching that of light. The only way in which his postulates can both be true is if space and time are not 'absolute'.

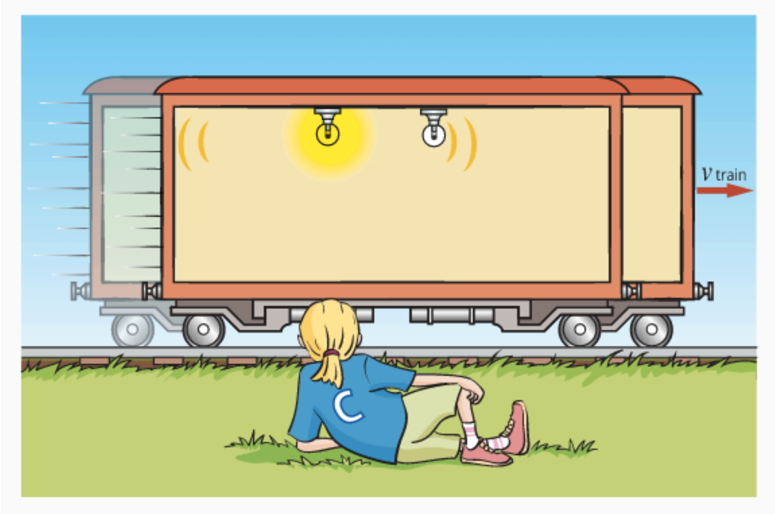
Einstein's Gedanken Train

Einstein discussed a simple thought experiment to illustrate the consequences of accepting the two postulates. It involves a ***train moving at a constant velocity***. ***Angela and Bill have boarded*** Einstein's train to help us with our experiments and ***Clare is still (stationary)*** outside on the platform.

This train has a flashing light bulb set right in the centre of the carriage. Angela and Bill are watching the flashes of light as they reach the front wall and back walls of the carriage, respectively. They are not surprised to find that the ***flashes reach the front and back walls at the same time***.



Outside, Clare is watching the same flashes of light. Einstein's interest was in when Clare saw the flashes hit the end walls.



The light has to reflect from the walls and then travel to Clare before she can 'see' it hit the wall, but the simplifying assumption made is that she makes the appropriate calculations to determine when the light 'actually' hit the walls - she ignores the ***look-back time***.

If Clare sees the light travelling at the same speed in the forward and back direction, ***she will see the light hit the back wall first***. This is because that wall is moving towards the light, whereas the front wall is moving away from the light and so the light will take longer to catch up to it.

Physics file

In discussing relativity, there are often situations where we want to know when light reaches a certain point, like the walls of the train. We (or our intrepid train travellers) only 'see' the light reach a point after it has reflected from that point and returned to our eyes. This extra time is called the *look-back time*. We need to calculate the look-back time and subtract it from the observed time to find the actual time to reach the wall. To avoid confusion, we will assume that our observers always do this and only quote the actual time.

This is all against the principles of Newtonian physics.

- Angela and Bill saw the light flashes reach the ends of the carriage at the ***same time***.
- Clare saw them reach the walls at ***different times***.

The idea that two events that are simultaneous for one set of observers are not simultaneous for another is a conundrum.

(Remember - Clare did not see the flashes at different times because of some sort of ***look-back time delay*** - that has been taken into account.)

This is often referred to as a ***lack of simultaneity*** - simultaneity means that a person would normally expect that, if one set of observers see two events happen simultaneously, they would expect any others to also see them simultaneously.

Simultaneity And Spacetime

How is it possible that two events that are simultaneous to one set of observers are not simultaneous to another?

Einstein said that the only reasonable explanation for this is that ***time itself is behaving strangely***. The amount of time that has elapsed in one frame of reference is not the same as that which has elapsed in another.

Because time (which has one dimension) seems to depend on the frame of reference in which it is measured, and a frame of reference is just a way of defining three-dimensional space, clearly time and space are somehow interrelated. Physicists call that ***four-dimensional relationship - spacetime***.

Special relativity is all about spacetime.

This was a profound shock to the physicists of Einstein's time. Many refused to believe that time was not the constant and unchanging quantity that it had always been assumed to be, and certainly always seemed to be.

And to think that it might 'flow' at a different rate in a moving frame of reference was too mind-boggling for words. That could mean that if a person went for a train trip, their ***clocks would go slow*** and they should come back ***younger*** than those who stayed behind.
This is what Einstein postulated, but younger by something much less than a microsecond - rather difficult to notice.

CONSEQUENCES OF SPECIAL RELATIVITY

There are ***three consequences*** of Einstein's special theory of relativity.

- ***Time Dilation***

If Einstein is correct that the speed of light is constant for each observer, regardless of their frame of reference, then one can ***compare the period of time for a light event*** in two frames of reference.

Example

Angela and Bill are travelling in a Gedanken space ship, while Clare observes from a space station. By observing a photon travelling from the light bulb in the space ship to the floor, one can determine the time taken for the event by dividing the distance travelled by the speed of light.

Angela and Bill see the light travelling the height of the space ship at the speed of light, so:

$$\Delta t_{\text{Angela + Bill}} = \frac{d}{c}$$

At the same time Clare, watching from the space station, sees the same event. The photon travels from the light bulb to the floor at the speed of light, but during that time the ***space ship has moved forward some distance*** determined by the velocity of the space ship and the period of time for the event. ***So Clare sees the photon travel at an angle to the ground***.

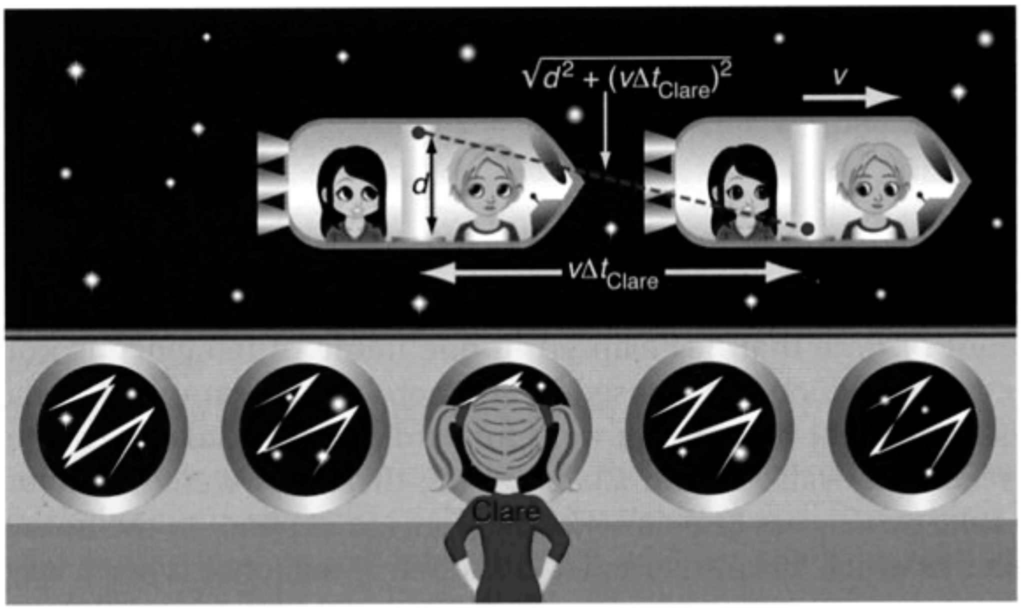


Figure 6.31

A photon event as seen by Clare. The photon has to travel a greater distance at the same speed of light, therefore the event takes a longer time.

The time that the photon takes to travel from the bulb to the floor is:

$$\Delta t_{Clare} = \frac{\sqrt{d^2 + \left(v\Delta t_{Clare}\right)}}{c}$$

The extra distance that the photon must travel in Clare's frame of reference means that the ***same event takes a longer period of time to occur***.

In fact, every event observed in Angela and Bill's frame of reference by Clare will occur over a longer period of time - the time taken for the second hand on Angela's watch to move will take longer, the time it takes for Bill to draw a breath will take longer.

The faster the space ship travels, the longer an event takes to occur, when viewed from a stationary frame of reference.

If Angela and Bill were riding a photon, how long would an event take when travelling at the maximum speed possible - the speed of light? The answer is - forever!

They would appear to Clare to be ***frozen in time***, for Angela and Bill everything else would also appear frozen in time, and ***it would appear to take no time to travel from one end of the Universe to the other***.

> ***Time dilation*** is when an event is timed in an inertial (constant non-zero) frame of reference by an observer in another inertial (stationary) frame of reference, and it will occur over a longer period of time than it does for the same event timed by an observer in the moving frame of reference.

To generalise Einstein's equation for time dilation, the symbol ***t*** is used to represent the dilated time that a stationary observer measures using a stationary clock, for an event that the observer sees occurring in a moving frame of reference. The symbol ***Δt*** is also used to represent this dilated time measured by a stationary observer. The symbol $\boldsymbol{t_0}$ is then the time that passes on the moving clock, which is also known as the ***proper time***. Proper time can also be represented by the symbol ***Δt′*** .

The factor that the proper time is multiplied by is given the symbol γ, so that:

$$\boxed{\mathbf{t = \gamma t_0}}$$

$$\text{where} \qquad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

The physicist H. A. Lorentz first introduced the factor γ in an attempt to explain the results of the Michelson–Morley experiment, so it is often known as the ***Lorentz factor***.

Example 1

Assume *Gedanken* conditions exist in this question. A stationary observer on Earth sees a very fast car passing by, travelling at $2.50 \times 10^8\ \mathrm{m\ s^{-1}}$. In the car is a clock on which the stationary observer sees $3.00\ \mathrm{s}$ pass. Calculate how many seconds pass by on the stationary observer's clock during this observation. Use $c = 3.00 \times 10^8\ \mathrm{m\ s^{-1}}$. 3 parts

Thinking	Working
Identify the variables: the time for the stationary observer is t, the proper time for the moving clock is t_0 and the velocities are v and the constant c.	$t = ?$ $t_0 = 3.00\ \mathrm{s}$ $v = 2.50 \times 10^8\ \mathrm{m\ s^{-1}}$ $c = 3.00 \times 10^8\ \mathrm{m\ s^{-1}}$
Use Einstein's time dilation formula and the Lorentz factor.	$t = \gamma t_0$ $= \frac{t_0}{\sqrt{\left(1 - \frac{v^2}{c^2}\right)}}$
Substitute the values for t_0, v and c into the equation and calculate the answer t.	$t = \frac{(3.00)}{\sqrt{\left(1 - \frac{(2.50 \times 10^8)^2}{(3.00 \times 10^8)^2}\right)}}$ $= \frac{(3.00)}{(0.55277)}$ $= 5.43\ \mathrm{s}$

For normal speeds, the Lorentz factor is essentially 1.000. As the object approaches relativistic speeds, the factor becomes very significant.

v (m s^{-1})	$\frac{v}{c}$	γ
3.00×10^2	0000001	1.000000000
3.00×10^5	000100	1.0000005
3.00×10^7	0.100	1.005
1.50×10^8	0.500	1.155
2.60×10^8	0.866	2.00
2.70×10^8	0.900	2.29
2.97×10^8	0.990	7.09
2.997×10^8	0.999	22.4

Example 2 - Long-lived Muons

In the Earth's atmosphere, high-energy cosmic rays interact with the nuclei of oxygen atoms 15 km above the surface of the Earth to create a cascade of high-velocity sub-atomic particles.

One of these particles is a ***muon***, which is unstable.

The mean lifetime of a stationary muon, as measured by an atomic clock, is 0.000 002 196 s (2.196 μs). The muons created by cosmic radiation typically travel at 99.97% of the speed of light, so at this speed, Newtonian physics would predict that a muon would travel about 659 m.

$$\begin{aligned} s &= v\Delta t \\ &= (0.9997)(3\times10^{8})(2.196\times10^{-6}) \\ &= 659 \text{ m} \end{aligned}$$

Given that, after 10 lifetimes, you can expect there to be essentially no muons remaining, then after a distance of 6.58 km, or at a height of about 8.42 km above the surface of Earth, you would expect that no muons would be detected.

However, you find that muons created by cosmic radiation are actually detected at the surface of the Earth. This means that the fast-moving muons have existed for a much longer period of time than they should have. A muon that strikes the surface would have existed at least 22.8 times its predicted lifespan as a stationary muon.

From the table above, the muon would be travelling at least 0.999c - hence it exists for longer than its normal mean ifetime.

Time dilation provides the explanation to this unusual observation.

Length Contraction

In the previous section, a photon travelling from the top of the space ship to the floor was used to compare the period of time for an event. It was deliberately set up in the spaceship so that this light path, of distance *d*, was ***perpendicular*** to the velocity. The reason for this was that we could expect distances in this perpendicular direction to be unaffected, but this may not be the case for distances in the direction of travel.

Einstein showed that while perpendicular distances are unaffected, ***relative motion affects lengths in the direction of travel***.

Einstein showed that as a person observes a fast-moving object, its length or the distance it travels in a certain time, will appear shorter than the distance seen when the object is stationary in the observer's reference frame.

It will be ***shorter by the same factor by which time appears to be extended*** and so Clare would see the length of the space ship contracted when compared to the length as measured by Angela and Bill.

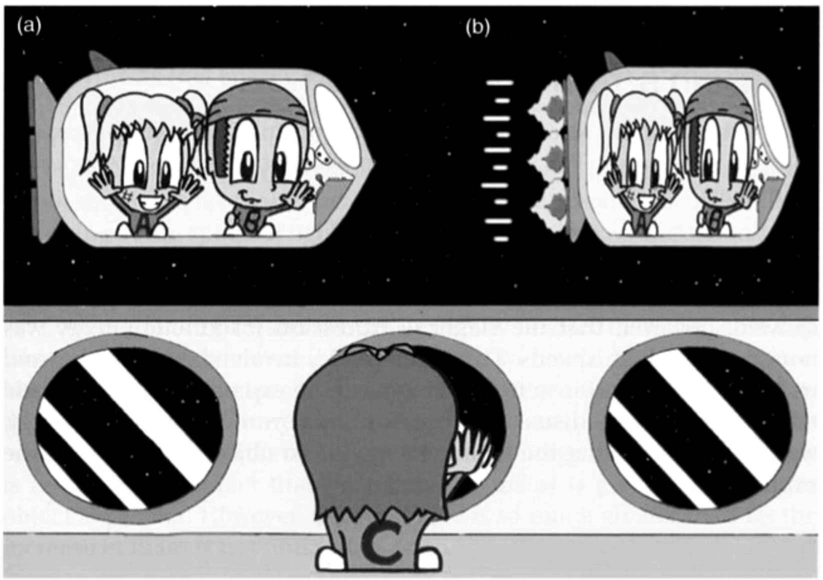


Figure 6.32
(a) Clare watches as Angela and Bill prepare for blast off. (b) As Angela and Bill speed past, Clare sees their rocket ship foreshortened, but the other dimensions remain the same.

Remember that ***length contraction occurs only in the direction of travel***, not in any perpendicular direction. To Clare, Angela and Bill's spaceship will appear foreshortened, but its width and height will remain unaltered.

These two simple notions - length contraction and time dilation - are at the core of the special theory of relativity. Together they tell Physicists about spacetime - the four-dimensional world that humankind inhabits.

Einstein's length contraction equation shows that an object with a proper length of $\boldsymbol{l_0}$, when measured at rest, will have a shorter length $\boldsymbol{l}$, parallel to the motion of its moving frame of reference when measured by an observer that is in a stationary frame of reference.

Length contraction can be represented as:

$$\boxed{l = \frac{l_0}{\gamma}}$$

This becomes:

$$\boxed{l = l_0\sqrt{1-\frac{v^2}{c^2}}}$$

Example

Assume *Gedanken* conditions exist in this question. A stationary observer on Earth sees a very fast car travelling by at $2.50 \times 10^8 \text{m s}^{-1}$. When stationary, the car is 3.00 m long. Calculate the length of the car as seen by the stationary observer. Use $c = 3.00 \times 10^8$ m s^{-1}.	3 parts
Thinking Identify the variables: the length for the stationary observer is l, the proper length of the moving car is l_0 and the velocities are v and the constant c.	Working $l = ?$ $l_0 = 3.00$ m $v = 2.50 \times 10^8$ m s^{-1} $c = 3.00 \times 10^8$ m s^{-1}
Thinking Use Einstein's length contraction formula and the Lorentz factor.	Working $l = \frac{l_0}{\gamma}$ $l = l_0\sqrt{\left(1-\frac{v^2}{c^2}\right)}$
Thinking Substitute the values for l_0, v and c into the equation and calculate the answer l.	Working $l = (3.00)\sqrt{\left(1-\frac{(2.50 \times 10^8)^2}{(3.00 \times 10^8)^2}\right)}$ $= (3.00)(0.553)$ $= 1.66$ m

Relativistic Mass and Momentum

Calculations involving time and distance get a bit rubbery when one considers relativistic effects like time dilation and length contraction. It would not be surprising, therefore, to hear that ***momentum also displays relativistic effects***.

If a rocket ship is travelling at 99% of *c*, why can't it simply turn on its rocket motor and accelerate up to 100%, or more, of *c*?

A full answer to this question was not given in Einstein's original 1905 paper on relativity. Some years later, he showed that as the speed of a rocket ship approaches *c*, its momentum increases as one might expect, but this is not reflected in a corresponding increase in speed.

While the force of the rocket engine increases the momentum, near the speed of light the increase in momentum results in ***smaller and smaller increases in speed***. So no amount of impulse can accelerate the rocket ship to the speed of light.

One interpretation of this strange behaviour of momentum is that ***it appears that the inertial mass of the rocket ship is increasing.***

Although Einstein preferred to stay with relativistic momentum, ***it is helpful to think in terms of the mass of the rocket ship increasing as its speed approaches* c**.

Hence, in terms of Newton's law, one can simply say that as the speed increases, the force required to produce a given acceleration increases towards infinity, because the mass approaches infinity.

Thus, no matter how much force is applied or work done, ***a rocket ship (or anything else) can never reach the speed of light***. In fact ***only things without mass*** can travel at the speed of light, like ***photons***.

The relativistic momentum is given by:

$$\boxed{p = \gamma p_0}$$

This becomes:

$$\boxed{p = \frac{p_0}{\sqrt{1-\frac{v^2}{c^2}}}}$$

Example

Assume *Gedanken* conditions exist in the following questions.

(a) Calculate the momentum, as seen by a stationary observer, provided to a rocket ship with a rest mass of 1000 kg, as it goes from rest up to a speed of $0.990c$. — 3 parts

Thinking	Working
Identify the variables: the rest mass is m_0, and the velocity of the rocket ship is v.	$\Delta p = ?$ $m_0 = 1000 \text{ kg}$ $v = 0.990c$
Use the relativistic momentum formula.	$p = \gamma m_0 v$
Substitute the values for m_0 and v into the equation and calculate the answer p.	$p = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} m_0 v$ $= \frac{1}{\sqrt{1-\frac{0.990^2 c^2}{c^2}}}(1000)(0.990 \times 3.00 \times 10^8)$ $= \frac{1}{\sqrt{1-0.990^2}}(1000)(2.97 \times 10^8)$ $= (7.08881)(1000)(2.97 \times 10^8)$ $= 2.11 \times 10^{12} \text{ kg m s}^{-1}$

(b) If twice the relativistic momentum from part (a) is applied to the stationary rocket ship, calculate the new final speed of the rocket ship in terms of c. — 3 parts

Thinking	Working
Identify the variables: the rest mass is m_0, and the relativistic momentum of the rocket ship is p.	$p = 2 \times (2.11 \times 10^{12})$ $= 4.21 \times 10^{12} \text{ kg m s}^{-1}$ $m_0 = 1000 \text{ kg}$ $v = ?$
Use the relativistic momentum formula, rearranged.	$p = \gamma m_0 v$ $= \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} m_0 v$ $v = \frac{p}{m_0\sqrt{\left(1+\frac{p^2}{m_0^2 c^2}\right)}}$
Substitute the values for m_0 and p into the rearranged equation and calculate the answer v.	$v = \frac{p}{m_0\sqrt{\left(1+\frac{p^2}{m_0^2 c^2}\right)}}$ $= \frac{(4.21 \times 10^{12})}{(1000)\sqrt{\left(1+\frac{(4.21\times 10^{12})^2}{(1000)^2(3.00\times 10^8)^2}\right)}}$ $= \frac{(4.21 \times 10^{12})}{(1000)\,(14.07)}$ $= 2.99 \times 10^8 \text{ m s}^{-1}$ $= 0.997c$

Relativistic Addition of Velocities

No two objects can have a relative velocity greater than *c*.

e.g. A spacecraft moving at 0.9*c* fires a missile at 0.8*c* forwards and relative to it.

According to classical Physics, a stationary observer sees this as 1.7*c*, which is impossible.

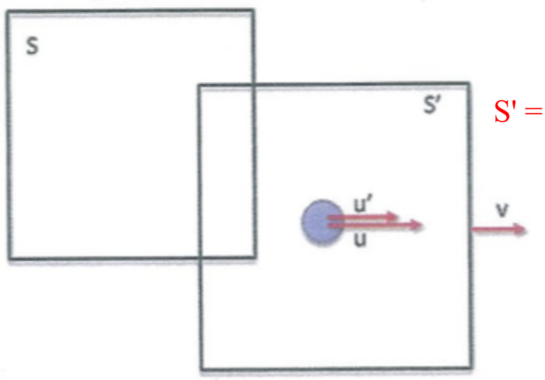
Velocities must transfornm according to the Lorentz transformation, leading to a non-intuitive result called ***Einstein velocity addition***.

(See Ch. 8.4 - Relativistic addition of velocities.)

Solution to the "problem"

Consider two observers in relative motion to each other and both are observing the motion of the missile.

S = stationary reference frame



S' = moving reference frame

v = velocity of moving reference frame

u' = velocity of missile in S' frame

u = velocity of missile relative to S frame

The equation describing *u* is given by:

$$\boxed{u = \frac{u' + v}{1 + \frac{u'v}{c^2}}}$$

where

u = velocity of the moving object in the statonary frame
u' = velocty of the moving object in the moving frame
v = velocity of the moving reference frame
c = speed of light

In this situation: u' = 0.8*c*, v = 0.9*c*.

i.e.

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

$$= \frac{0.8c + 0.9c}{1 + \frac{(0.8c)(0.9c)}{c^2}}$$

$$= \frac{1.7c}{1.72}$$

$$= \underline{0.988c}$$

Example

Two spaceships are moving apart from each other, A at 0.7*c* and B 0.5*c* in the opposite direction. What is the velocity of A as seen by an observer in B?

Consider B as the stationary reference frame.

Take the direction of B as +ve.

v = -0.7c
u = 0.5c
u' = velocity of A relative to B

$$u' = \frac{u - v}{1 + \frac{uv}{c^2}}$$

$$= \frac{0.5c - (-\ 0.7c)}{1 + \frac{(0.5c)(-\ 0.7c)}{c^2}}$$

$$= \frac{1.2c}{1.35}$$

$$= \underline{0.889c}$$

∴ <u>A is receding at 0.889c from B</u>

Note If the direction of A is taken as +ve, u' = - 0.889c.

The -ve sign means that A is receding from B.

The result is the same.

Einstein's Equation

As the momentum of an object increases, so does the energy. The classical relationship between the two can be written as:

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}pv$$

As v approaches relativistic speeds near c, the kinetic energy, as well as the momentum, will increase towards infinity.

Einstein showed that the classical expression for kinetic energy was not correct at high speeds.

The mathematics involved is beyond this course but Einstein, working from the expression for relativistic momentum and the usual assumptions about work, forces and energy,
was able to show that the kinetic energy of an object was given by the expression:

$$E_k = (\gamma - 1)m_o c^2$$

where E_k = the kinetic energy of an object corrected for relativistic effects (J).
γ = the Lorentz factor, which is the factor that converts rest mass to relativistic mass.
m_o = the mass of an object when it is at rest in our frame of reference (kg).
c = the speed of light (3.00×10^8 ms^{-1}).

Einstein's expression can be rewritten as:

$$E_k = (\gamma - 1)m_o c^2$$
$$= \gamma\, m_o c^2 - m_o c^2$$
$$\Rightarrow \quad \gamma\, m_o c^2 = E_k + m_o c^2$$

Einstein interpreted this expression as being an expression for the ***total energy***, of the object.

$$\boxed{E_{total} = E_{kinetic} + E_{rest}}$$

$$\Rightarrow \quad E_T = \gamma\, m_o c^2 = E_k + m_o c^2$$

Now as γm_o is equal to the relativistic mass (m), the total energy can simply be written as:

$$E_T = mc^2 = E_k + m_o c^2$$

where $E_k = {}^1/_2\, mv^2$

Einstein wondered - what is the significance of the rest energy, the m_oc^2 term?

As this energy depends directly on the mass of the object, it appears to be energy associated with mass, and so mass, he realised, was a form of energy.

Physicists now understand that this equation shows that mass and energy are totally interrelated.

Example 1

A cup of hot water is ever so slightly heavier than the same cup when cold. The difference is far less than one could possibly measure, but it is.

The heat means the molecules have more kinetic energy, and as the total energy is the sum of the rest and kinetic energy, the total energy (mc^2) is greater.

This is reflected in the fact that m (relativistic mass) is greater, so a hotter object is heavier.

Since m_oc^2 is so much greater than E_k, the increase in mass is not noticeable.

Example 2

Calculate how much more massive a 200 g cup of boiling water is than the same cup of iced water, given that the specific heat capacity of water is 4.18×10^3 J kg^{-1}K^{-1}. 5 parts

Thinking Identify the variables for calculating the increase in energy (in this case, the increase in the internal energy of the water).	*Working* $m = 0.200$ kg $c_w = 4.18 \times 10^3$ J kg^{-1} K^{-1} $\Delta T = 100$ K
Thinking Use the appropriate formula (in this case, the specific heat formula).	*Working* $Q = mc_w\Delta T$
Thinking Substitute the values into the equation and calculate the energy (in this case, Q).	*Working* $Q = (0.200)(4.18 \times 10^3)(100)$ $= 8.36 \times 10^4$ J
Thinking Use the total energy formula.	*Working* $E = mc^2$
Thinking Equate the energy values (if necessary), substitute the values for E and c into the equation, and calculate the answer m.	*Working* $m = \frac{E}{c^2} = \frac{Q}{c^2}$ $= \frac{(8.36 \times 10^4)}{(3.00 \times 10^8)^2}$ $= 9.29 \times 10^{-13}$ kg (This increase in mass is very tiny.)